

FRE-GT-9743 Assignment 3

Jianing(Jay) Yao

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Bond Future: Bond future is the exchange version of bond forward, which has a more complex mechanism due to standardization. The main feature of bond future is:

- market observable future price F
- delivery date of the future T_d
- the underlying basket of bonds $\{B_x\}_{x \in \mathcal{X}}$ that are potentially to be delivered¹
- due to the fact that each bond has different coupon and maturity, conversion factor K is introduced for normalization
- the cheapest-to-deliver (CTD) bond changes over time until the delivery date

1. Compute normalize forward price: Given a 10-year bond that has EXACTLY 9 years to maturity, coupon $c = 4\%$, semi-annual pay, spot price $S = \$100$.

- 1) compute its 3-month forward price BF given market repo level $r_{\text{mkt}} = 2\%$;
- 2) compute the conversion factor K , defined as the clean price of the bond assuming a constant yield of 6%, divided by 100;
- 3) the normalized forward price is BF/K .
- 4) we need to perform this calculation for all bonds in the basket, intuitively, do you see what conversion factor does? (*Hint: pay attention to accrued interest.*)

¹The bonds in $\mathcal{X}$ cannot differ significantly in maturities, i.e., 5Y to maturity bond and 25Y to maturity bond won't be in the same delivery basket.

2.Cheapest-To-Deliver: The future price is given by:

$$F = \min_{x \in \mathcal{X}} \frac{BF_x(S, r_x, T_d)}{K_x} \quad (1)$$

The market quotes F , which means we can back out r_x , this is called *implied repo* for bond x .

1) Suppose future price is \$115.3670, what is the implied repo for the bond in *question 1* ?

2) Say, you have computed implied repo for all bonds in the basket $\{r_x^{\text{imp}}\}_{x \in \mathcal{X}}$, and you know roughly the 3M general collateral repo rate in the market r_{mkt} , how do you decide which bond is the cheapest to deliver as of today ?

(Hint: the exchange needs to deliver the bond, what they would do if they observe $r_{\text{mkt}} \neq r_{\text{imp}}$?)

3.Scenario Analysis: Can you predict CTD changes, i.e., given the market condition (or what you think will be), what CTD might be N-day later ? and why do you care about which bond to deliver in the end ?

Computational Part

Please follow the structure in the FixedIncomeLib.

github link:

<https://github.com/Jerry0108-SJ/FRE-GY-9743-Assignments-3.git>